

Faculty of Science and Engineering

Computer Science

CS-210 Concurrency

May/June 2024

Time Allowed: 2 Hours

Do NOT turn over your question paper until instructed to do so.

Exam Paper Information

Answer all questions.

Special Instruction(s)

none

Specific Items

Dictionaries – Candidates may only refer to the English and Welsh language dictionaries available

Calculators – Candidates may NOT use a calculator

Open Book – This is NOT an Open Book Exam

Contact Details

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NB. The content of this exam paper has been thoroughly checked, however if you think you have spotted an error, please raise your hand and report to an invigilator. Unless it has a significant impact on the paper the error will NOT be corrected in the exam venue.

Errors not corrected in the exam should be noted in your answer booklet and you should continue to answer the examination paper to the best of your ability as it is written. It will be noted by the module coordinator and the Board of Examiners when it considers your results.

1. Give short explanations of the following concepts:
 - (a) Concurrency. **[2 marks]**
 - (b) Shared memory. **[2 marks]**
 - (c) Semaphore. **[2 marks]**
 - (d) `notifyAll()` in Java. **[2 marks]**
 - (e) Multiverse in Java. **[2 marks]**
2. A shared object is used to count events. It has three operations: `start`, `event` and `tally`, where
 - `start` initialises/resets the counter to 0;
 - `event` increments the counter;
 - `tally` returns the counter value if it is non-zero or blocks until it is non-zero otherwise.
 - (a) Write the FSP code that models the object. (You may assume that the number of events will remain small in the modelling, e.g. 5.) **[5 marks]**
 - (b) Write a Java monitor that implements the desired behaviour. **[5 marks]**
3. Threads in Java:
 - (a) Show how to create and start threads in Java (use code). **[3 marks]**
 - (b) Explain and compare `Thread` and `Runnable` in Java. **[4 marks]**
4. Deadlocks
 - (a) Describe the Coffmann conditions for deadlock. **[4 marks]**
 - (b) Brother and sister fight over a pair of running shoes. Sometimes this results in each one getting one shoe. To stop the fight, their father takes the left shoe from who got it and gives it to the other. Why is this not a deadlock? Relate to the conditions in (a). **[2 marks]**

5. Amdahl's law is formulated as $L(k, n) \leq \frac{t_i}{t_o(k, n)} = \frac{1}{\frac{1}{k}t_s + \frac{1}{n}t_p}$, where the performance gain L depends on the sequential improvement factor k and the number of cores n .

(a) What are $t_o(k, n)$ and $t_i = t_o(1, 1)$? **[2 marks]**

(b) Derive the law. **[5 marks]**

(c) A parallelised programme spends 75% of its time in its parallel part. Use Amdahl's formula to compute the maximum speedup bound possible with 5 cores. **[2 marks]**

(d) In the scenario described in (c), what is the upper bound on the speedup regardless of how many cores are used? **[1 mark]**

6. Consider the following FSP programme.

$P = (a \rightarrow Q \mid b \rightarrow Q),$
 $Q = (c \rightarrow P).$

(a) Draw the corresponding LTS. **[1 mark]**

(b) The LTSA tool will not report any progress violations for this programme. Why not? **[2 marks]**

(c) Explain how to investigate adverse conditions in LTSA with the above example. **[2 marks]**

7. Explain what the image below indicates in an RAG.



[2 marks]

End of Paper